

BITBURG, Germany

WMO: 106100

Lat: 49.950N

Lon: 6.567E

Elev: 1227

StdP: 14.06

Time Zone: 1.00 (EUC)

Period: 82-93

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	12.6	17.7	4.8	7.4	14.4	9.0	9.1	20.0	29.6	45.1	26.3	41.4	9.7	60	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.0	82.8	65.9	80.4	65.7	76.9	63.4	68.1	79.3	66.5	77.4	64.8	74.6	6.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	94.7	72.1	62.6	88.9	70.9	60.9	83.5	69.5	33.3	79.8	31.8	77.0	30.4	74.5	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.7	19.8	17.6	DB	11.7	90.6	7.0	3.5	6.7	93.1	2.6	95.2	-1.3	97.2	-6.4	99.8	(4)
(5)			WB	10.5	71.7	6.7	2.1	5.6	73.2	1.7	74.4	-2.0	75.6	-6.9	77.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	33.6	33.1	40.4	46.0	53.8	58.9	64.3	63.9	57.3	49.6	40.0	36.3
	DBStd	12.80	8.31	8.01	6.58	6.52	6.59	6.06	5.92	5.88	5.49	6.13	6.31	6.59
	HDD50	2297	509	475	304	154	36	4	0	0	5	84	302	424
	HDD65	6283	974	894	763	570	348	201	90	91	236	478	749	889
	CDD50	1639	0	1	7	34	155	270	442	432	224	70	3	1
	CDD65	148	0	0	0	0	2	16	67	58	5	0	0	0
	CDH74	1348	0	0	0	2	31	140	594	517	64	0	0	0
	CDH80	304	0	0	0	0	3	14	154	127	6	0	0	0
Wind	WSAvg	7.4	9.5	8.3	8.6	7.4	6.7	6.5	6.2	5.8	6.4	7.3	7.6	7.9
Precipitation	PrecAvg	30.2	2.8	2.2	2.0	1.9	2.6	2.8	2.8	2.6	2.4	2.6	2.6	3.1
	PrecMax	39.7	7.8	5.9	5.6	4.8	5.0	6.7	7.6	5.7	5.4	7.2	5.2	8.5
	PrecMin	22.2	0.3	0.2	0.3	0.0	0.5	0.8	0.9	0.4	0.7	0.9	0.2	0.4
	PrecStd	4.3	1.5	1.3	1.3	1.0	1.2	1.5	1.4	1.3	1.4	1.3	1.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	50.4	55.4	64.4	70.2	77.3	82.1	88.1	89.2	80.6	70.2	55.6	54.0
		MCWB	47.2	47.4	52.2	56.1	60.4	67.1	67.8	66.6	63.7	60.7	52.7	50.5
	2%	DB	48.4	50.2	58.7	66.0	73.5	78.6	84.2	82.8	75.3	64.7	53.5	51.4
		MCWB	46.0	45.4	48.5	53.1	59.8	64.4	66.5	65.7	63.6	58.3	50.5	48.2
	5%	DB	46.5	46.5	53.7	61.2	70.2	73.8	80.6	80.2	71.3	61.1	51.7	48.1
		MCWB	45.0	43.2	47.1	50.8	57.7	62.5	66.2	65.5	60.7	56.7	49.1	46.1
	10%	DB	44.6	43.1	50.1	57.3	66.4	70.1	76.9	75.5	67.6	58.6	48.6	45.0
		MCWB	42.7	40.4	45.0	48.5	56.2	60.9	63.6	63.1	59.2	54.6	46.5	43.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.7	49.2	53.1	57.7	63.1	70.2	70.7	69.7	67.0	61.9	54.0	51.2
		MCDB	50.3	54.4	61.3	68.8	74.3	77.8	83.0	83.0	75.1	67.7	55.3	53.0
	2%	WB	46.4	46.2	49.9	54.3	60.8	66.4	68.3	67.5	64.3	59.6	51.1	48.5
		MCDB	47.7	49.9	56.5	63.6	71.0	74.9	80.7	79.3	72.4	64.3	53.0	50.5
	5%	WB	44.7	43.2	47.8	51.6	58.8	64.0	66.7	65.8	62.0	57.2	49.4	46.0
		MCDB	46.2	45.8	53.0	59.6	68.0	71.8	78.1	77.2	68.9	60.9	51.5	47.6
	10%	WB	42.5	40.7	45.5	49.6	57.1	61.6	64.9	64.2	60.0	55.1	47.0	43.9
		MCDB	43.8	43.0	49.7	56.3	65.4	69.1	75.2	74.1	66.2	58.5	48.5	45.0
Mean Daily Temperature Range	5% DB	MDBR	7.8	10.1	12.8	15.6	16.5	15.3	18.0	18.0	15.4	12.3	8.9	7.4
		MCDBR	10.0	14.0	19.5	21.8	22.7	21.4	24.2	24.1	22.3	16.4	11.2	9.7
		MCWBR	9.2	11.3	12.8	12.7	12.7	11.8	11.5	11.2	13.2	11.1	10.0	9.2
	5% WB	MCDBR	9.8	12.7	17.1	19.8	20.3	19.1	22.1	21.3	18.9	15.6	9.9	8.8
MCWBR		9.4	11.0	12.2	12.4	12.1	11.4	11.1	10.8	12.3	10.9	9.3	8.5	
Clear-Sky Solar Irradiance	taub	0.303	0.328	0.359	0.390	0.399	0.401	0.399	0.395	0.367	0.350	0.324	0.300	
	taud	2.436	2.387	2.322	2.268	2.274	2.288	2.287	2.327	2.385	2.429	2.440	2.427	
	Ebn at Noon	237	257	267	270	272	272	271	266	262	245	225	219	
	Edh at Noon	20	26	33	39	40	40	40	36	31	25	20	17	
All-Sky Solar Radiation	RadAvg	250	477	847	1335	1574	1747	1655	1419	1061	613	283	193	
	RadStd	38	90	129	164	158	159	180	144	129	59	34	29	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page